

Resilient decentralized communication for unstructured peer-to-peer networks

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Abstract

Recent crises, natural disasters, and large-scale protests emphasize how people rely heavily on online social networks and instant messaging platforms to communicate, while organizations use those platforms to coordinate aid efforts. During crises, consolidated communication systems can become strained, partially or wholly destroyed, censored, or otherwise monitored. Free distributed alternatives to consolidated networks are peer-to-peer networks, which distribute the power and content control over many, equal peers. Previous studies evaluated algorithms, metrics and performance of structured overlays of p2p networks. This paper evaluates decentralization metrics of unstructured p2p networks, by providing empirical and simulated insights to their performance and resilience. We propose design considerations to support resilient overlay communication amid clusterization.

1 Introduction

The decentralized administration of the Internet is one of the basic enablers of the rapid growth of the Internet as a whole, thanks to the autonomous nature of different network systems [16]. Different Internet Service Providers (ISPs) play a critical role in the functionality of the Internet, as they connect their end customers (Internet users) to the Internet backhaul. The power dynamic of this system inherently gives control over shaping consumption patterns to the ISPs: consolidation of power at the hands of a few gives the ISPs the unique ability to control, throttle, or monitor the flow of information, in addition to enabling oppressive practices such as censorship and surveillance of end users. Governmental control of the telecommunication infrastructure often leads to the repression of communities during times of political unrest [2, 9], severely limiting the possibility for social actors to dissent and organize themselves. In addition to policial repression engaged by the government, physical disruptions can also limit free speech. Most recently, communication disruptions due to airstrikes on main fiber optic lines, and the depletion of fuel for generators, have severely limited the information flow to the population of Gaza since the start of the Israel-Hamas war in October 2023, halting humanitarian coordination, and leaving the affected population entirely cut off from life-saving information and assistance [8, 27].

Together, the aforementioned practices have led to a growing interest in peer-to-peer (p2p) messaging platforms, decentralized social networks [3, 26], and distributed ledger-based applications [6, 22]. To guarantee free speech in disaster zones, during protests, and for citizens living under oppressive regimes, we decide to focus on unstructured p2p networks, as they inherently exhibit a

higher level of resilience to topology changes and churn, in comparison to structured p2p networks. Structured p2p networks usually require an expensive randomized approach to establish resilient topologies, as they need to spread information across random peers to prevent dependence on any single peer [12, 14, 21]. Additionally, iterative diffusion of the information causes data plane overhead [14], which increases the message propagation latency [12, 13]. Popular blockchains, which use a structured p2p overlay, employ concentration of services in a few highly available nodes, poses a threat to resilience despite a high reliability of those nodes, as a single network partition can prevent the entire network from communicating [11]. Other famous structured p2p networks include systems as Chord [29], Kademlia [21], and the InterPlanetary File System (IPFS) [4]. Unstructured p2p networks, on the other hand, are more suitable for highly-dynamic alternative communication methods, often necessary to guarantee free decentralized communication means, as they usually do not distribute central information amongst the participating nodes. Most notably, the rise of unstructured p2p messaging platforms like Firechat [28], Briar [25], bitchat [18], and Amigo [17], have proven that there is a growing shift in user interest patterns away from traditional Centralized Social Networks towards diverse p2p social networks.

After exploring the different types of p2p overlay networks, we discuss the choice of the communication protocol, which dictates the limited physical device communication radius. We chose to focus on a stable protocol that most modern devices support and the populace is most likely to use in disaster zones: Bluetooth, and more specifically Bluetooth Low Energy (BLE) [5]. We decided to focus on Bluetooth Low Energy (BLE) as it is more power efficient, over classic Basic Rate/Enhanced Data Rate (BR/EDR), as BLE conserves device battery life, it allows users to communicate over short ranges over a longer period of time than comparable communication protocols e.g. WiFi Direct [10].

This paper provides the design space for resilient unstructured p2p networks, based on sparsity of the underlying topologies.

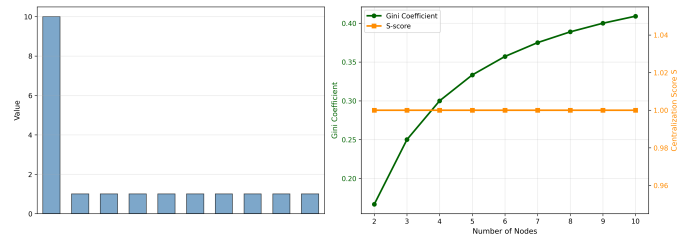


Figure 1: Sparse topology

The property we used to measure resilience and sparsity is the node degree, or the number of connections a node has to other peers in the network topology: an overall similarity between individual node degrees increases the resilience of a topology, as network partitions affecting a subset of the nodes will not result in a complete blackout of the network because many alternative paths still exist. To illustrate, figure 1 shows an increasingly sparse network topology, in which one node becomes increasingly dominant as it accumulates more network links, and more nodes poorly connected nodes are added. While the left plot simply shows the distribution of values, the right plot compares the different resilience scores that this paper uses: the Gini coefficient in green, and the Centralization score $\mathcal{S}$ in orange.

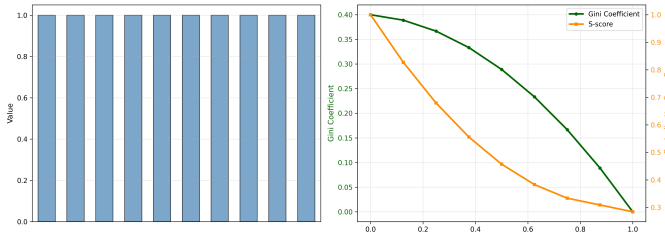


Figure 2: Dense topology

An increasingly dense topology can be observed in figure 2, that illustrates the change in resilience metrics as the topology becomes equally distributed. The value at the start of this plot represents an extremely unequal distribution, where one node dominates all network links, and as the plot progresses on the X axis, the topology gradually reaches a state of perfect equilibrium where all nodes are equally connected. It is noteworthy to observe that the Gini coefficient at the end of this plot reaches a minimum value of 0, indicating perfect equality, where the $\mathcal{S}$ score does not: this is because the Gini coefficient is calculated by computing the difference in individual concentration of value in the network, and when all those differences are zero, the Gini coefficient also reaches zero. The Gini measurement focuses on the **overall** inequality distribution, where the $\mathcal{S}$ score focuses on the **top** dominant values, which makes it inherently sensitive to outliers.

In this research paper, we focus on the problem of designing resilient unstructured p2p overlay networks using BLE. To do that, we conducted an experiment using modern devices to collect real-world empirical data on the stochastics of the BLE signal propagation over different distances, and use the collected empirical evidence to parameterize simulations of unstructured p2p networks. We measure the resilience [15, 23], and theoretical channel capacity of those simulated networks, and rationalize about the design of resilient local p2p overlay networks using our sparsity metrics. For this, in section 2, we discuss the Bluetooth protocol, and the state of the art of sparsity and resilience metrics that capture resilience and centralization of network topologies. Next, Section 3 describes the setup of the experiment testbed that captures the BLE measurements, presents the empirical results from the testbed which we then use as parameters for the simulations, and closes with detailing the simulation setup 3.4. Following the setup section,

section 4 evaluates the results from the simulations, concluded by the discussion on section 5.

2 State of the Art

In this section we discuss the state-of-the-art of related works upon which this research builds, and deals with the question: how can centralization metrics be used to measure the resilience of unstructured p2p networks?

Traditionally using standard IP packet forwarding, which requires an end-to-end connection between peers, p2p networks can opportunistically use short-range wireless p2p communication protocols to Store, Carry, and Forward [7] messages while the Internet infrastructure is unavailable, as presented with Twimight [20]. Twimight and most unstructured peer-to-peer networks usually rely on flooding and gossip for content dissemination, as these networks do not store information about the cluster membership. Simulating an unstructured p2p network, our simulations similarly utilize flooding and gossip for neighbor discovery and content dissemination. The topologies and stochastics of the simulations are captured, processed, and their resilience measured using centralization metrics. To bring these concepts into the context of global-scale networks, we adapt centralization and sparsity metrics to measure the resilience of different network topologies at scale.

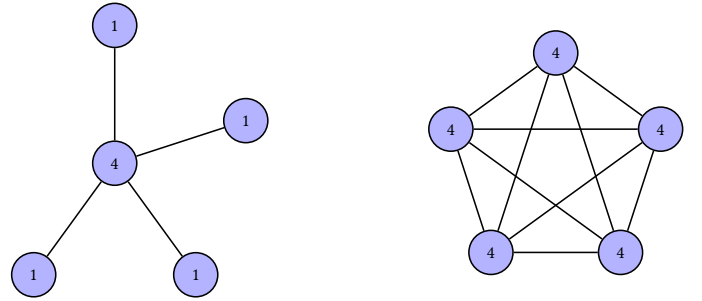


Figure 3: Sparse vs. dense topologies

To compute the different centralization metrics of our different topologies, we extract a (weighted) vector of node degrees from each topology, and compare the values of the different metrics derived from it. The first centralization metric used in this paper is the common reference metric: the L_0 norm, which simply counts the number of non-zero elements in a vector. The L_0 norm is calculated using the following formula:

$$L_0 = \frac{\sum_{i=0}^n \begin{cases} 1 & \text{if } a_i > 0 \\ 0 & \text{otherwise} \end{cases}}{n}$$

where a_i is the degree of node i , and n is the total number of nodes.

The second centralization metric used in this paper is the Gini coefficient [23], which measures the overall inequality of a distribution, or sparsity: a small number of components in a distribution hold a large proportion of the overall value. Different measures of sparsity have been introduced and ranked according to fulfilling desirable criteria, that sparsity measures should logically satisfy [23]. Amongst the most common methods to measure sparsity,

the Gini coefficient outperformed all other measurements considered, satisfying all criteria. The Gini coefficient, a metric commonly associated with the Human Development Index to measure income inequality around the world, is calculated as follows:

$$G = \frac{\sum_{i=0}^n \sum_{j=0}^n |a_i - a_j|}{2n \sum_{i=0}^n a_i}$$

Where a_i is the degree of node i , and n is the total number of nodes.

Lastly, the third metric we used is the Centralization Score $\mathcal{S}$ [15], which is derived from the Herfindahl-Hirschmann Index (HHI). The HHI index is used to measure market competition in US antitrust laws. In their work, Habib et al. study patterns of providing multiple web services in different countries around the world, and demonstrate how regions with higher centralization around a smaller number of providers are more susceptible to outages. The Centralization Score $\mathcal{S}$ is defined as follows:

$$\mathcal{S} = \sum_{i=0}^n \left(\frac{a_i}{C}\right)^2 - \frac{1}{C}$$

where C is the total number of edges (i.e. network links), a_i is the degree of node i , and n is the total number of nodes. Centralization metrics can be used to measure the resilience of unstructured p2p networks by capturing the inequality in the distribution of node degrees, and comparing different network topologies: we argue, that topologies with an overall more similar distribution of node degrees are inherently more resilience.

3 Setup

3.1 Testbed

Our testbed consists of a set of Raspberry Pi 3B+ devices, using the built-in BLE supporting chip Cypress CYW43455, and external power supplies of 5v/3.4A to ensure stable operation and mobility. The Raspberry Pis were flashed with a lightweight 64-bit Linux port (Raspberry Pi OS Lite based on Debian Trixie), upon which the standard Bluetooth stack was installed using BlueZ 5.66. During the experiment, one Raspberry Pi acted as the transmitting node, continuously sending BLE advertisements at a fixed interval of 20ms + $\mathcal{X}$, where $\mathcal{X}$ is a random variable uniformly distributed between 0 and 10ms, representing the delay added to avoid collisions with other devices using the same advertising interval [5, 24]. The other pi was the scanner, running a Python script that uses the Bleak library to continuously scan for a window set to one minute, looking for BLE advertisements originating from the other pi.

In BLE, two types of scanning are possible: passive scanning, where the scanner only listens for ads, and active scanning, where upon receiving an ad, the scanner also sends extended scan requests to the advertiser node for additional data. In this experiment, passive scanning is used. BLE ads form the bottleneck of BLE communications, as ads are used to discover and connect to neighbours in local networks. The two devices were placed at increasing distances from each other, where the advertiser node was kept static and the receiver node was moved. During scanning, the receiver node recorded the RSSI of all ads received from the transmitting node, as well as the transmitting power (tx) and the timestamp at reception. Each distance step was repeated twenty times to capture enough data points for statistical significance. The distances measured were

from 0 meters to 10 meters in increments of 5 meter, and then from 10 meters in increments of 10 meters, until no more ads were received for most runs of that distance. The experiment was divided to two parts: in the first part, the distance points up to 30 meters were evaluated in an urban backyard; in the second part, the distance points from 40 meters were evaluated in an olympic athletics field. Both parts of the experiment were conducted in an open level environment with minimal surrounding interference, clear direct line of sight between the nodes, and minimal multipath effects. To measure the BLE signal coverage stochastics and how the signal propagates in free space, we compared the Received Signal Strength Indicator (RSSI) of BLE advertisements at the receiver node across increasing distances from the transmitter node.

3.2 Empirical Insights

The Cumulative Distribution Function (CDF) plot of the RSSI values received at different distances is shown in figure 4. As expected, the figure shows a general decreasing trend of the RSSI values, in correlation with the increase in distance. The jittery nature of the CDF curves at distances ≥ 40 meters can be attributed to the lower number of samples collected at those ranges, which leads to a higher variability in the RSSI measurements. The Probability Density Function (PDF) plot and a time series plot of the RSSI values for each distance can be found in the appendix, figures 11 and 12.

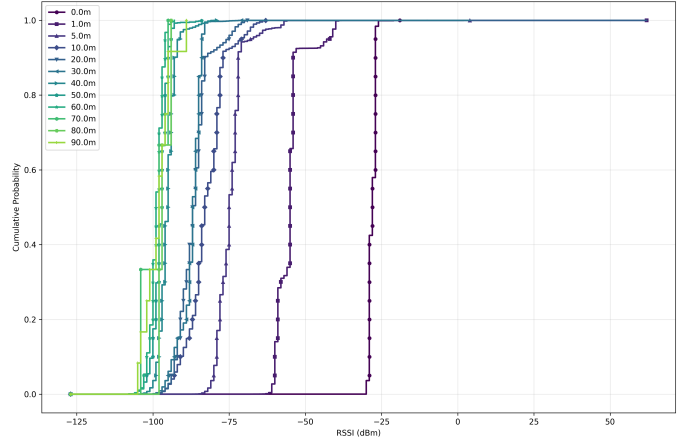


Figure 4: RSSI CDF

Figure 5 shows the advertisement delivery success probability at different distances, or the percentage of ads received at the receiver node, depending on two estimates of the number of ads transmitted. The figure showcases two subplots: the top plot calculates the ratio between the count of ads received at distance X , and the count of ads received at 0 meters, where the two devices were placed right next to each other. This plot assumes that the count of ads received at 0 meters is the same as the count of ads that were actually sent. The bottom plot estimates the number of ads that the transmitter node sent, by dividing the length of time of the scanning interval by the advertising interval of 20ms and adding to it the expected value of the random variable $\mathcal{X}$: 5ms. It is trivial to see that as the distance increases, the advertisement delivery success probability decreases,

following a roughly exponential decay. No ads were received for any run for distances more than 90 meters, so the experiment was concluded at that point.

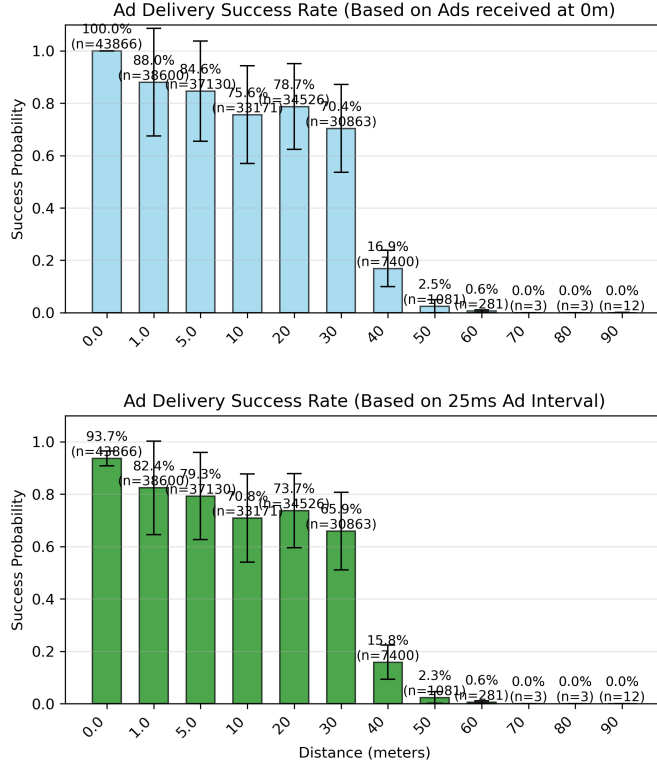


Figure 5: Advertisement Delivery Success Probability

Next, table 1 shows summary statistics of the RSSI values received at different distances: the mean RSSI values presented at the table were used to parameterize the simulation signal propagation decay with distance. Throughout all distances, the same transmitting power was recorded and thus not displayed in the table, which hints at the default transmitting power of a Raspberry Pi 3b+: 12 dBm.

3.3 Simulations

We used the The ONE simulator [19] to simulate an opportunistic exchange of messages between nodes communicating over BLE, using the empirical results from the testbed to model the signal path loss with distance. The simulation was repeated for 25 runs per configuration, using an Ubuntu 22.04 with an Intel(R) Xeon(R) Gold 6326 CPU @ 2.90GHz with 64 cores, and 247 GB of RAM. To run the simulations we used parallelization, spawning 60 parallel jobs. Using a static movement model, every node remains immobile and attempts to send the same amount of messages to all other nodes in the network. To mimic the scenario of message propagation in disaster zones, we used an epidemic router to flood messages across the network. In an epidemic routing scheme, every node sends all messages, including messages it relays, to all of its neighbors, and each neighbor recursively repeats this iterative process until the

Table 1: RSSI Statistics by Distance

Distance (m)	Mean RSSI (dBm)	Std Dev	Num Samples
0.0	-28.03	1.21	43,866
1.0	-55.14	4.62	38,600
5.0	-74.58	4.77	37,130
10.0	-82.47	6.21	33,171
20.0	-86.54	5.05	34,526
30.0	-87.45	3.55	30,863
40.0	-95.21	2.76	7,400
50.0	-97.50	2.44	1,081
60.0	-98.82	2.10	281
70.0	-99.00	3.74	3
80.0	-96.33	1.70	3
90.0	-98.25	4.23	12

final destination node for each message is reached. For every simulator run, we used either exclusively intra-cluster communication to simulate clique-based short-range cluster communication, or inter-cluster communication to simulate long-range communication: In the former, a host generates 20 messages to all other hosts in its cluster, while at latter every host generates 20 messages to all other hosts in the network. We used the typical BLE MTU size of 247 bytes for all messages. A simulator run ends when all generated messages are either delivered or discarded. For the signal path loss model, we used empirical data presented in table 1, and interpolated for distances between the empirical measurements using the normal logarithmic path loss model laid out in equation 6 of [1]. The path loss model is defined as follows:

$$RSSI_d(\text{dBm}) = A - 10 * \alpha * \log_{10}\left(\frac{d}{d_0}\right)$$

where $RSSI_d$ is the received signal strength indicator at distance d between source and destination nodes, d_0 is the distance at the nearest lower or equal reference point, A is the RSSI at the nearest lower or equal reference distance d_0 (as measured by the testbed), and α is the path loss exponent. We wanted to evaluate the performance of the Bluetooth Low Energy running on 1M PHY mode, which means a modulation rate of 1Mbps at ≤ 1 meter. To calculate the possible channel bitrate cap at a specific range, we used the Shannon-Hartley theorem to derive the theoretical capacity based on known BLE channel properties. A large 400×400 meter L-shape building hall was generated, split to a grid of 12 equal-sized cells of 100×100 meters each assigned to a cluster consisting of six hosts, for a total of 72 hosts. We randomly placed each host within the cell to which its cluster was assigned. To generate the random node positions, a random number generator was used. We fed the random generator with a seed that was mathematically derived from the communication range, message size, and optionally by the simulation run number, to ensure result reproducibility. The communication range is a configurable parameter which limits the radius of communication of the network interface that each node has: for simplification, a connection between two nodes can be established if they are within the communication radius of each other, and have direct line of sight. A connection does not churn during a

simulation run: once a connection is established it is maintained until the end of the simulation run.

Using the results of the BLE signal propagation from the testbed, we created simulations for the different communication radii tested in the pi experiment. A simulation run configuration consists of a set of parameters, including the message generation rate, the maximum communication radius of each node, the maximum number of parallel connections (max node degree) each node can establish and maintain, and the communication mode. The same exact topology generated for each communication radius was repeated for maximum node degrees between 1-20, each for 25 runs, randomizing the topologies with each run. In doing so, this shuffled the order of the hosts picked for message generation, which we used to prevent introducing bias by the order of the hosts, and created more variance in the message distance travelled. This was done to elicit the effects of the maximum node degree and communication range on timely message delivery.

3.4 Model Insights

Figure 6 shows the distance distribution of all messages created, aggregated over all simulation runs, communication ranges, and maximum node degrees tested, split by the communication mode. It is evident that the intra-cluster communication mode had a generally higher overall delivery probability, possibly attributed to a lesser strain on the individual hosts' message buffers.

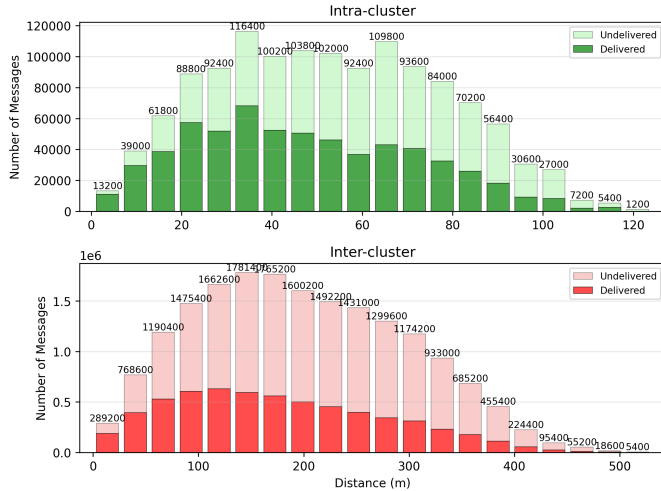


Figure 6: Message Distance Distribution

To better understand the topology, figure 7 shows the distance matrix between hosts to the left, supplemented by two topological matrices, one for each communication mode. The topological matrices have a black cell at position (i, j) if host i is connected to host j . Next, randomized topologies were created for each configuration, to assess how the maximum node degree affects the message latency, centralization, and sparsity metrics of unstructured p2p networks. This was done to evaluate how different topologies with the same communication radius and maximum node degree induce variance in the different metrics

4 Evaluation

Finally, to evaluate the performance of different topologies by their centralization metrics, we use the Gini and Centralization Score $\mathcal{S}$ and plot them against the maximum constrained node degree.

Figure 8 shows how the different centralization metrics introduced in section 2 correlate with an increase in the maximum constrained node degree. An extended version of this plot, comparing the aforementioned centralization metrics to the L_0 norm also in the intra-cluster communication mode, can be found in the appendix, figure 14. As evident from this plot, the Gini coefficient has a higher variance in comparison to the $\mathcal{S}$ score, which appears to be more stable as the maximum constrained node degree increases. This

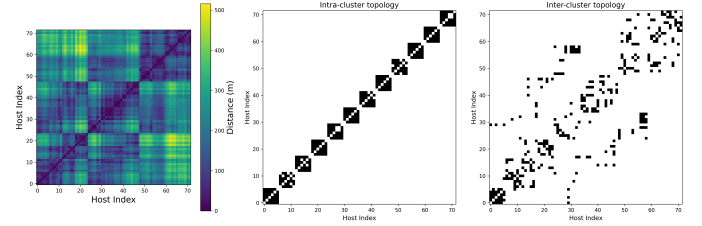


Figure 7: Topology and distance metrics for communication radius 90m, constrained maximum node degree 5

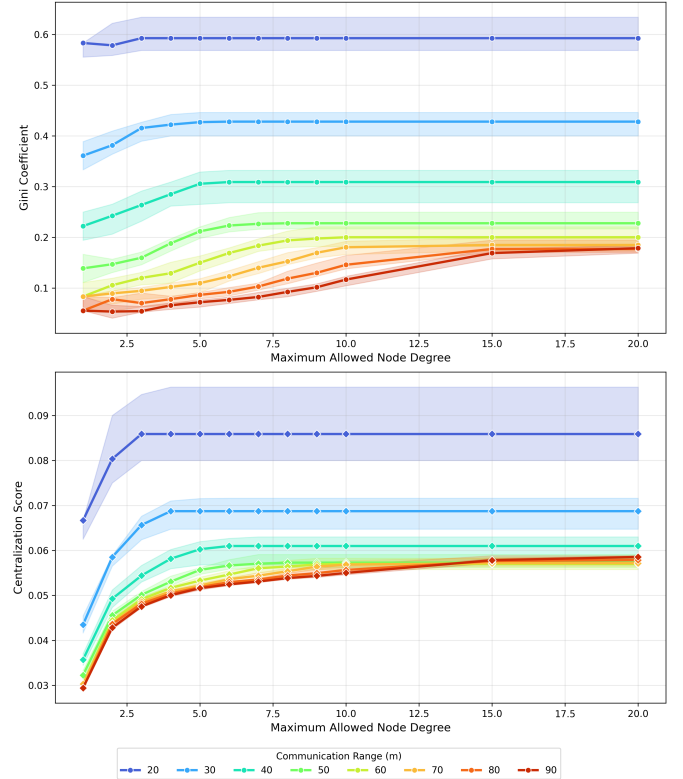


Figure 8: Centralization vs. max node degree - inter-cluster communication

indicates that the Gini coefficient is more appropriate to measure resilience.

Then, in figure 9 we plot the different centralization metrics against the maximum node degree for the maximum communication radius of 90m, to understand how the peer degree impacts the end-to-end message latency at a typical BLE 1M PHY range. We bin all the messages found for the 90m communication range to three distance categories: short (≤ 120 m), medium (120 – 220m), and long (> 220 m) distances. We can see an initial increase in order of magnitude in the latency, as the maximum node degree increases from two to three, but as we approach higher degrees of around ten, the mean latency actually decreases, especially in long-distance messages. Additionally, the long tail of the short distance messages indicates most direct links are short ranged, as expected.

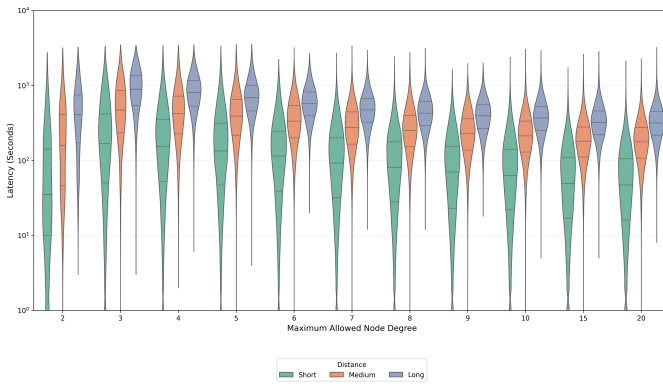


Figure 9: Latency vs. Node degree at 90m range- inter-cluster communication

Figure 10 shows how centralization correlates with mean message delivery latency. The top subplot showcases the Gini coefficient on the X axis, while the bottom subplot showcases the $\mathcal{S}$ Score. The Gini plot generally is able to extract more information, as the Gini coefficient is sensitive to the overall distribution of data, where the $\mathcal{S}$ -Score is more sensitive to extreme values: the Gini score ranges from about 0.1 (the left extreme point of the X axis), until almost 1.0 (at the extreme right); this tells us, that it is able to capture more variance in the data. In comparison, the bottom $\mathcal{S}$ -Score subplot generally saturates very quickly, offering a breadth of less than 0.1 of variance for most communication radius. In Gini plot, a clear relationship is visible for most communication radii between an increase in the centralization scores, and an increase in the end-to-end message latency. The exception lies with the 90m communication radius, which exhibits a negative correlation between centralization and latency: the latency decreases with an increase in centralization. This could be attributed to the fact that nodes in this communication radius were able to establish direct links with their intended recipients, which reduces the amount of hops required to deliver a message.

5 Discussion

The results shown at figure 8 and in more depth at figure 14 in the appendix, indicate that increasing the maximum constrained node

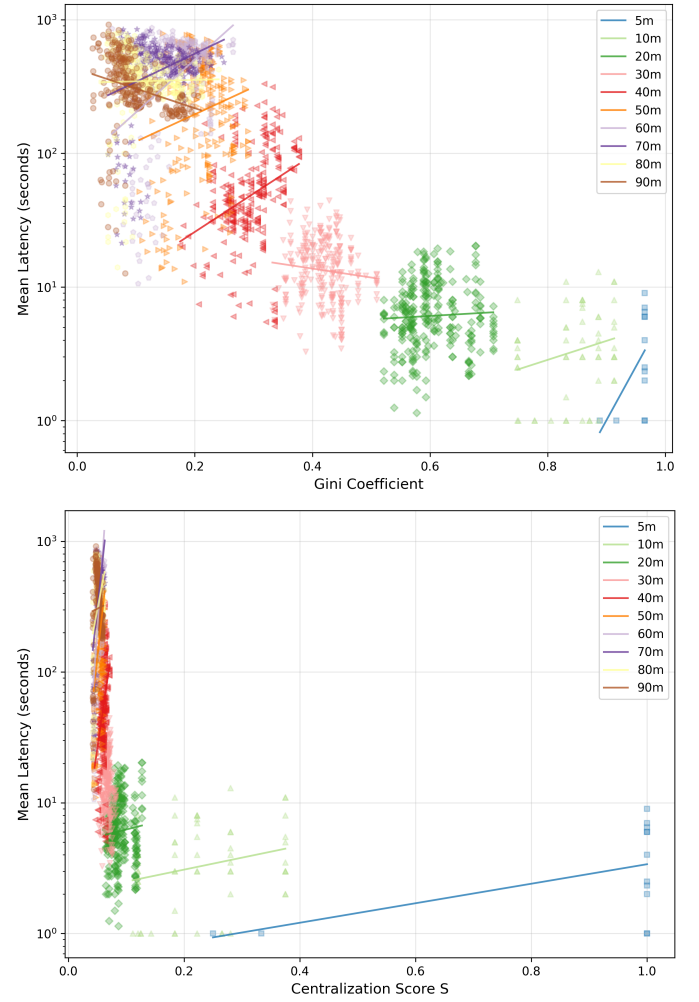


Figure 10: Latency vs. Centralization - inter-cluster communication

degree generally increases the inequality and centralization of the resulting topology, as measured by both the Gini index and the Centralization Score $\mathcal{S}$. This effect is more pronounced in higher communication ranges, as visible with the stagnation in the $\mathcal{S}$ -score for medium communication ranges 50-80m, around the maximum allowed node degree of 10. The Gini coefficient exhibits less of a stagnation effect, which indicates it is more sensitive to changes in the topology than the $\mathcal{S}$ -Score. This is due to the fact that the Gini coefficient depends on the entire node degree distribution, whereas the $\mathcal{S}$ -Score depends mostly on the top dominant nodes. Because of the inherent focus of the $\mathcal{S}$ -Score on the top dominant nodes, it saturates quickly as the maximum allowed node degree increases, as the topologies exhibit less of a star-like structure. It is however noteworthy to observe that both metrics are usable, albeit for different goals: the Gini coefficient is more appropriate to measure resilience as quantified by an even distribution of node degrees, while the $\mathcal{S}$ -Score is more appropriate to detect single points of failure. To design a topology that is more resilience, it

is therefore important to consider both the intended communication range, centralization rating, and the physical distribution of nodes: if the nodes are geographically dispersed and a higher communication range is needed, the established network might result in a sparsely-populated network which is susceptible to centralization effects such as vulnerability to outages. Furthermore, in unstructured p2p networks, employing gossip to disseminate information without consideration of the peer degree, can lead to a very high message bloat that may lead to communication bottlenecks and reduced resilience, as nodes with a high peer degree become quickly overwhelmed with messages to process, and can no longer effectively communicate themselves. Lastly, the cost of communication-intensive algorithms, such as ones used by epidemic routers or structured p2p networks that are both characterised by a high messaging overhead, should be considered when designing unstructured p2p networks, as they can quickly saturate the limited channel capacity of BLE links, leading to increased message latency.

6 Future Work

Due to the inherent variance in the BLE protocol and high interference demonstrated on wireless channels commonly used in the urban settings used for the empirical experiment, some RSSI values should be collected again, as visible from jittery subplots in the PDF plot 11: for example, the 60m and 90m plot exhibit repeating jumps in their density values, in comparison to the smoothness in lower ranges (e.g. 10, 20, 30m ranges). This we attribute to the low number of received packets at those distances (sometimes below 100 packets 5), which makes the RSSI distribution more sensitive to outliers. The RSSI values we have provided for the ranges that are complete and stable, are mostly sufficient to derive the necessary path loss parameters for the simulation, as they exhibit the expected power-law distribution and are mostly consistent. Furthermore, we were not able to have fine-grain control over exactly when the advertisements were sent, because the BlueZ Bluetooth communication stack and the underlying hardware controller handle the advertisement scheduling internally to prevent antenna overheating. To best mitigate this, we set the advertisement interval window to the minimum possible values of to achieve the highest consistent frequency of advertisements sent. We were however able to prove that the advertisements were not actually being consistently received at the configured interval window of 20ms using a Bluetooth sniffer on a mobile phone, which measured the advertisement intervals. Therefore, another prospect for this project in the future, would be to use a custom-made BLE firmware on a microcontroller platform, such as the Nordic nRF52 series, to have more control over the advertisement scheduling and transmission power. This would additionally allow us to measure the effect setting the transmission power has on the signal propagation distance, which can allow us to model the resilience of p2p networks built using different transmission power levels.

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7 Appendix

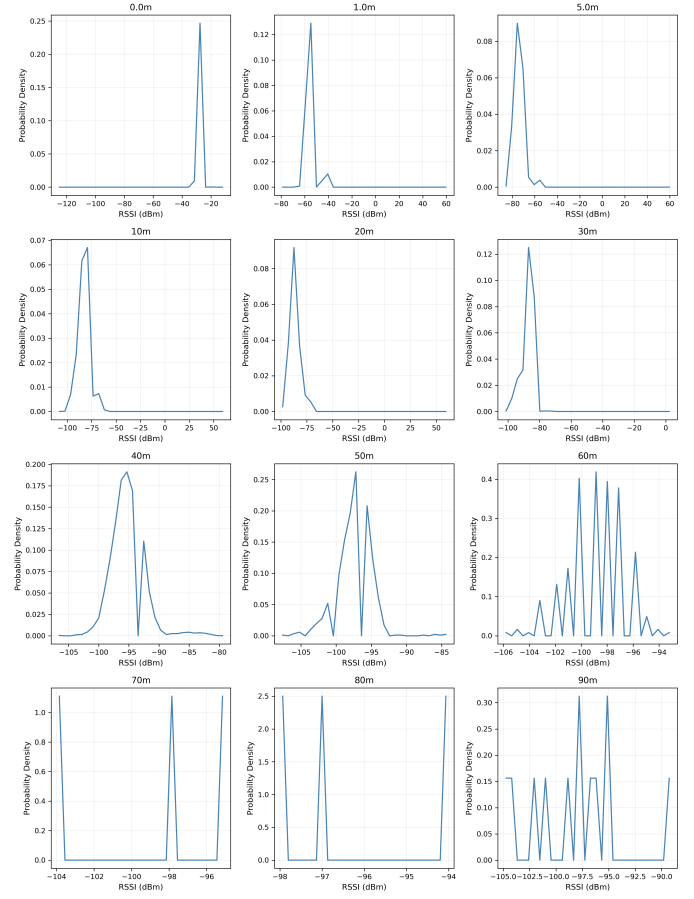


Figure 11: RSSI PDF

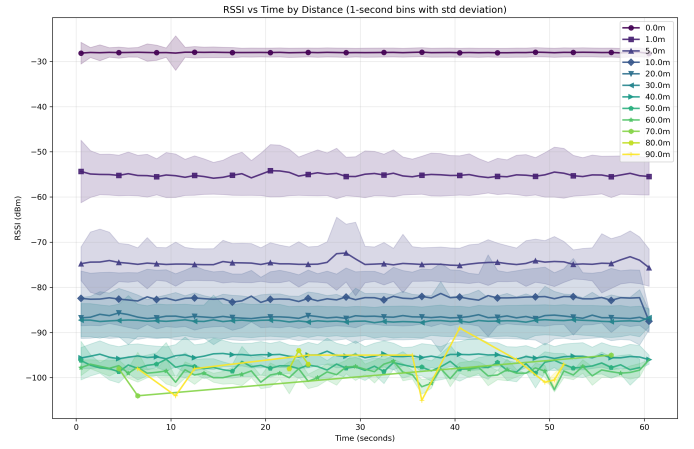


Figure 12: RSSI time series

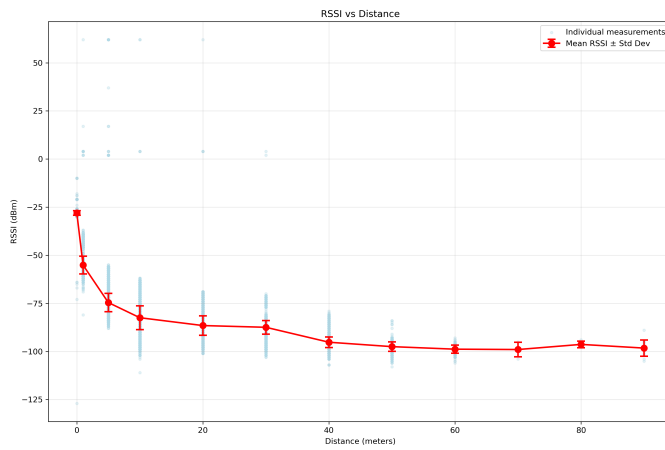


Figure 13: Mean RSSI at different distances

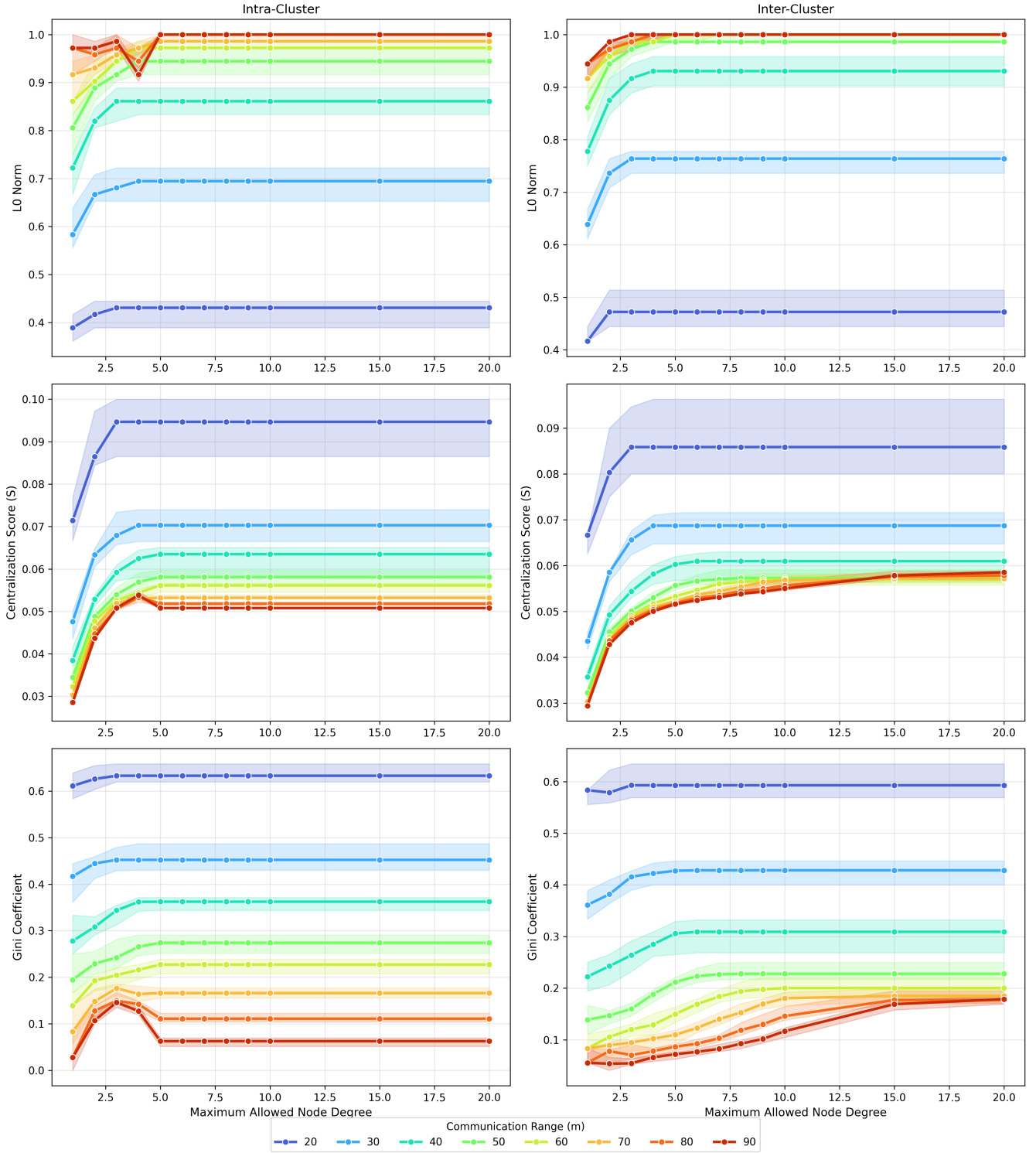


Figure 14: Centralization metrics vs maximum node degree